

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287684

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SHINSHO; Kohei

SEMICONDUCTOR DEVICE

Abstract

A semiconductor device includes a first conductivity type semiconductor layer having a first and second surfaces, device and outer regions, a channel region of a second conductivity type in a surface layer portion of the first surface in the device region, an emitter region of a first conductivity type in a surface layer portion of the channel region, a gate electrode at the first surface in the device region, the gate electrode facing the channel region across a gate insulating film, a collector region of a second conductivity type in a surface layer portion of the second surface in the device region, an inner cathode region of a first conductivity type in the surface layer portion of the second surface in the device region, and an outer cathode region of a first conductivity type in the surface layer portion of the second surface in the outer region.

Inventors: SHINSHO; Kohei (Kyoto-shi, JP)

Applicant: ROHM CO., LTD. (Kyoto-shi, JP)

Family ID: 63037367

Assignee: ROHM CO., LTD. (Kyoto-shi, JP)

Appl. No.: 19/218649

Filed: May 27, 2025

Foreign Application Priority Data

JP	2017-012264	Jan. 26, 2017
----	-------------	---------------

Related U.S. Application Data

parent US continuation 17693957 20220314 parent-grant-document US 12336275 child US 19218649

parent US continuation 16827187 20200323 parent-grant-document US 11309310 child US 17693957

Publication Classification

Int. Cl.: **H10D84/60** (20250101); **H10D8/00** (20250101); **H10D12/00** (20250101); **H10D62/10** (20250101); **H10D62/13** (20250101); **H10D62/17** (20250101); **H10D64/00** (20250101); **H10D84/80** (20250101)

U.S. Cl.:

CPC **H10D84/617** (20250101); **H10D8/00** (20250101); **H10D12/441** (20250101); **H10D12/461** (20250101); **H10D12/481** (20250101); **H10D62/104** (20250101); **H10D62/106** (20250101); **H10D62/107** (20250101); **H10D62/108** (20250101); **H10D62/112** (20250101); **H10D62/142** (20250101); **H10D64/112** (20250101); **H10D84/811** (20250101); H10D62/393 (20250101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. patent application Ser. No. 17/693,957, filed Mar. 14, 2022, which is a Continuation of U.S. patent application Ser. No. 16/827,187, filed Mar. 23, 2020 (now U.S. Pat. No. 11,309,310), which is a Continuation of U.S. patent application Ser. No. 15/878,863, filed Jan. 24, 2018 (now U.S. Pat. No. 10,636,786), which claims priority to Japanese Patent Application No. 2017-012264, filed Jan. 26, 2017, the entire disclosures of each are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to a semiconductor device.

2. Description of the Related Art

[0003] FIG. 2 of US2010276727A1 discloses an RC-IGBT (Reverse Conducting-Insulated Gate Bipolar Transistor) serving as an example of a semiconductor device.

[0004] The RC-IGBT includes a semiconductor layer. A p type channel region is formed in a surface layer portion of a front surface of the semiconductor layer. An n type emitter region is formed in a surface layer portion of the channel region. A gate electrode electrically connected to the channel region across an insulating film is formed at the front surface of the semiconductor layer. A p type connector region and plural n type cathode regions are formed in a region of a surface layer portion of a rear surface of the semiconductor layer, the region facing the channel region.

SUMMARY OF THE INVENTION

[0005] One preferred embodiment of the present invention provides a semiconductor device that includes a semiconductor layer of a first conductivity type having a first principal surface on one side and a second principal surface on the other side, the semiconductor layer in which a device formation region and an outer region outside the device formation region are set, a channel region of a second conductivity type formed in a surface layer portion of the first principal surface of the semiconductor layer in the device formation region, an emitter region of a first conductivity type formed in a surface layer portion of the channel region, a gate electrode formed at the first principal surface of the semiconductor layer in the device formation region, the gate electrode facing the channel region across a gate insulating film, a collector region of a second conductivity type

formed in a surface layer portion of the second principal surface of the semiconductor layer in the device formation region, an inner cathode region of a first conductivity type formed in the surface layer portion of the second principal surface of the semiconductor layer in the device formation region, and an outer cathode region of a first conductivity type formed in the surface layer portion of the second principal surface of the semiconductor layer in the outer region.

[0006] One preferred embodiment of the present invention provides a semiconductor layer of a first conductivity type having a first principal surface on one side and a second principal surface on the other side, a channel region of a second conductivity type formed in a surface layer portion of the first principal surface of the semiconductor layer, an emitter region of a first conductivity type formed in a surface layer portion of the channel region, a gate electrode formed at the first principal surface of the semiconductor layer, the gate electrode facing the channel region across a gate insulating film, a collector region of a second conductivity type formed in a facing region facing the channel region in a surface layer portion of the second principal surface of the semiconductor layer, an inner cathode region of a first conductivity type formed in the facing region in the surface layer portion of the second principal surface of the semiconductor layer, and an outer cathode region of a first conductivity type formed in a non-facing region which is a region outside the facing region in the surface layer portion of the second principal surface of the semiconductor layer.

[0007] The above and other objects, features, and effects of the present invention will be made clear by the following description of the preferred embodiments with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a top view of a semiconductor device according to a first preferred embodiment of the present invention.

[0009] FIG. 2 is a sectional view taken along line II-II shown in FIG. 1.

[0010] FIG. 3 is a sectional view taken along line III-III shown in FIG. 1.

[0011] FIG. 4 is a plan view for explaining a structure on the second principal surface side of a semiconductor layer shown in FIG. 1.

[0012] FIG. 5 is a circuit diagram showing an electrical structure of the semiconductor device shown in FIG. 1.

[0013] FIG. 6 is a graph in which forward current-forward voltage characteristics of the semiconductor device shown in FIG. 1 are obtained by simulation.

[0014] FIG. 7 is a graph in which a peak forward surge current of the semiconductor device shown in FIG. 1 is obtained by simulation.

[0015] FIG. 8 is a plan view showing a first modification example of an outer cathode region.

[0016] FIG. 9 is a plan view showing a second modification example of the outer cathode region.

[0017] FIG. 10 is a plan view showing a third modification example of the outer cathode region.

[0018] FIG. 11 is a plan view showing a fourth modification example of the outer cathode region.

[0019] FIG. 12 is a plan view showing a first modification example of an inner cathode region.

[0020] FIG. 13 is a plan view showing a second modification example of the inner cathode region.

[0021] FIG. 14 is a plan view showing a third modification example of the inner cathode region.

[0022] FIG. 15 is a plan view showing a fourth modification example of the inner cathode region.

[0023] FIG. 16 is a view showing a modification example of a cathode region.

[0024] FIG. 17 is a schematic sectional view of a semiconductor device according to a second preferred embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0025] With the semiconductor device according to US2010276727A1, when a predetermined forward voltage is applied between the emitter region and the collector region in a state where a predetermined voltage is applied between the gate electrode and the emitter region, it is sometimes difficult for a forward current to flow between the emitter region and the collector region.

[0026] Therefore, in order to obtain a relatively high forward current, a relatively high forward voltage has to be applied. As a result, the forward voltage is deteriorated. The deterioration of the forward voltage can result in a snap-back phenomenon.

[0027] Thus, a preferred embodiment of the present invention provides a semiconductor device with which deterioration of a forward voltage can be suppressed.

[0028] One preferred embodiment of the present invention provides a semiconductor device that includes a semiconductor layer of a first conductivity type having a first principal surface on one side and a second principal surface on the other side, the semiconductor layer in which a device formation region and an outer region outside the device formation region are set, a channel region of a second conductivity type formed in a surface layer portion of the first principal surface of the semiconductor layer in the device formation region, an emitter region of a first conductivity type formed in a surface layer portion of the channel region, a gate electrode formed at the first principal surface of the semiconductor layer in the device formation region, the gate electrode facing the channel region across a gate insulating film, a collector region of a second conductivity type formed in a surface layer portion of the second principal surface of the semiconductor layer in the device formation region, an inner cathode region of a first conductivity type formed in the surface layer portion of the second principal surface of the semiconductor layer in the device formation region, and an outer cathode region of a first conductivity type formed in the surface layer portion of the second principal surface of the semiconductor layer in the outer region.

[0029] With this semiconductor device, even when a predetermined forward voltage is applied between the emitter region and the collector region in a state where a predetermined voltage is applied between the gate electrode and the emitter region, difficulty of a forward current in flowing between the emitter region and the collector region can be suppressed. Therefore, the deterioration of the forward voltage can be suppressed. Thus, a snap-back phenomenon due to the deterioration of the forward voltage can be suppressed.

[0030] One preferred embodiment of the present invention provides a semiconductor device that includes a semiconductor layer of a first conductivity type having a first principal surface on one side and a second principal surface on the other side, a channel region of a second conductivity type formed in a surface layer portion of the first principal surface of the semiconductor layer, an emitter region of a first conductivity type formed in a surface layer portion of the channel region, a gate electrode formed at the first principal surface of the semiconductor layer, the gate electrode facing the channel region across a gate insulating film, a collector region of a second conductivity type formed in a facing region facing the channel region in a surface layer portion of the second principal surface of the semiconductor layer, an inner cathode region of a first conductivity type formed in the facing region in the surface layer portion of the second principal surface of the semiconductor layer, and an outer cathode region of a first conductivity type formed in a non-facing region which is a region outside the facing region in the surface layer portion of the second principal surface of the semiconductor layer.

[0031] With this semiconductor device, even when a predetermined forward voltage is applied between the emitter region and the collector region in a state where a predetermined voltage is applied between the gate electrode and the emitter region, difficulty of a forward current in flowing between the emitter region and the collector region can be suppressed. Therefore, the deterioration of the forward voltage can be suppressed. Thus, a snap-back phenomenon due to the deterioration of the forward voltage can be suppressed.

[0032] The preferred embodiments of the present invention shall now be described in detail with reference to the attached drawings.

[0033] FIG. 1 is a top view of a semiconductor device 1 according to a first preferred embodiment of the present invention.

[0034] With reference to FIG. 1, the semiconductor device 1 includes a semiconductor layer 2 formed in a chip shape. The semiconductor layer 2 includes a first principal surface 3 on one side, a second principal surface 4 on the other side, and four side surfaces 5a, 5b, 5c, 5d connecting the first principal surface 3 and the second principal surface 4.

[0035] The first principal surface 3 and the second principal surface 4 are formed in a square shape in a plan view as viewed from a normal direction of the first principal surface 3 of the semiconductor layer 2 (hereinafter, simply referred to as “in the plan view”). The side surface 5a and the side surface 5b face each other. The side surface 5c and the side surface 5d face each other. A device formation region 6, an outer region 7, a withstand voltage retention region 8, and a scribe region 9 are set in the semiconductor layer 2. The device formation region 6 is also called the active region.

[0036] The device formation region 6 is a region where an IGBT (Insulated Gate Bipolar Transistor) is formed. The device formation region 6 is set to be spaced from a peripheral edge of the semiconductor layer 2 to an inner region of the semiconductor layer 2.

[0037] The device formation region 6 is set in a central region of the semiconductor layer 2 in the plan view. The device formation region 6 is set in a square shape having four sides parallel to the side surfaces 5a to 5d of the semiconductor layer 2 in the plan view.

[0038] The outer region 7 is set in a region on the outside of the device formation region 6. The outer region 7 is a region partitioning the device formation region 6 from the other region (withstand voltage retention region 8). The outer region 7 is set in a region between the peripheral edge of the semiconductor layer 2 and a peripheral edge of the device formation region 6. The outer region 7 is set in an endless shape (square ring shape) surrounding the device formation region 6 in the plan view.

[0039] The withstand voltage retention region 8 is set in a region on the outside of the outer region 7. The withstand voltage retention region 8 is a region for improving withstand voltage of the semiconductor device 1. The withstand voltage retention region 8 is set in a region between the peripheral edge of the semiconductor layer 2 and a peripheral edge of the outer region 7. The withstand voltage retention region 8 is set in an endless shape (square ring shape) surrounding the outer region 7 in the plan view.

[0040] The scribe region 9 is set in a region on the outside of the withstand voltage retention region 8. The scribe region 9 is a region where a cutting member such as a dicing blade passes through at the time of manufacturing. The scribe region 9 is set in a region between the peripheral edge of the semiconductor layer 2 and a peripheral edge of the withstand voltage retention region 8. The scribe region 9 is set in an endless shape (square ring shape) surrounding the withstand voltage retention region 8 in the plan view.

[0041] A front surface electrode 10 is formed above the first principal surface 3 of the semiconductor layer 2. The front surface electrode 10 includes a gate electrode 11, an emitter electrode 12, field plate electrodes 13, and an equipotential electrode 14. The gate electrode 11, the emitter electrode 12, the field plate electrodes 13, and the equipotential electrode 14 are electrically insulated by an insulating region 15 bordering those electrodes.

[0042] The gate electrode 11 is formed in the outer region 7. The gate electrode 11 includes a gate pad 16 and a gate finger 17. The gate pad 16 is formed along a central region of the side surface 5a in the plan view.

[0043] In this embodiment, the gate pad 16 is formed in a square shape in the plan view. The gate pad 16 crosses a region of the device formation region 6 and the outer region 7 and is drawn out from the outer region 7 into the device formation region 6.

[0044] The gate finger 17 is drawn out in a band shape from the gate pad 16, and partitions the device formation region 6 from three directions in the outer region 7. The gate finger 17 has a pair

of open ends **18**, **19** on the side surface **5b** side. More specifically, the gate finger **17** includes a first gate finger **20** and a second gate finger **21**.

[0045] The first gate finger **20** is drawn out in a band shape from an end portion of the gate pad **16** on the side surface **5c** side. The first gate finger **20** has the open end **18** on the side surface **5b** side. The second gate finger **21** is drawn out in a band shape from an end portion of the gate pad **16** on the side surface **5d** side. The second gate finger **21** has the open end **19** on the side surface **5b** side.

[0046] The emitter electrode **12** includes an emitter pad **22**, an emitter routing portion **23**, and an emitter connecting portion **24**.

[0047] The emitter pad **22** is formed in a C-shaped region partitioned by a peripheral edge of the gate pad **16** and a peripheral edge of the gate finger **17** in the plan view. The emitter pad **22** is formed in a C shape along the peripheral edge of the gate pad **16** and the peripheral edge of the gate finger **17** in the plan view.

[0048] The emitter pad **22** covers the substantially entire device formation region **6**. A peripheral edge of the emitter pad **22** crosses the region of the device formation region **6** and the outer region **7** and is drawn out from the device formation region **6** into the outer region **7**.

[0049] The emitter routing portion **23** is formed in the outer region **7**. The emitter routing portion **23** is drawn around in a band shape in a region on the outside of the gate finger **17**. In this embodiment, the emitter routing portion **23** is formed in an endless shape (square ring shape) surrounding the gate finger **17** in the plan view. The emitter routing portion **23** may be formed in an ended shape extending in a band shape along the gate finger **17**.

[0050] The emitter connecting portion **24** crosses a region between the pair of open ends **18**, **19** of the gate finger **17** from the emitter pad **22** and is connected to the emitter routing portion **23**. The emitter routing portion **23** is electrically connected to the emitter pad **22** via the emitter connecting portion **24**.

[0051] The IGBT formed in the device formation region **6** structurally includes an npn type parasitic bipolar transistor. When an avalanche current generated in a region outside the device formation region **6** flows into the device formation region **6**, the parasitic bipolar transistor is turned on. In this case, control of the IGBT may sometimes become unstable due to a latch-up.

[0052] Thus, in this embodiment, an avalanche current absorbing structure **25** that absorbs the avalanche current generated in the region outside the device formation region **6** is formed by the emitter electrode **12** including the emitter pad **22**, the emitter routing portion **23**, and the emitter connecting portion **24**.

[0053] In the avalanche current absorbing structure **25**, the avalanche current generated in the region outside the device formation region **6** is absorbed by the emitter routing portion **23**. The avalanche current is taken out from the emitter pad **22** via the emitter connecting portion **24**.

[0054] Thereby, turning-on of the parasitic bipolar transistor by an undesired current generated in the region outside the device formation region **6** can be suppressed. Therefore, a latch-up can be suppressed, so that stability of the IGBT can be enhanced.

[0055] The field plate electrodes **13** are formed in the withstand voltage retention region **8**. In FIG. **1**, the field plate electrodes **13** are shown by lines. In this embodiment, the plural (five in this case) field plate electrodes **13** are formed to be spaced from each other from the outer region **7** toward the scribe region **9**.

[0056] Each of the field plate electrodes **13** is drawn around in a band shape along the emitter routing portion **23**. In this embodiment, each of the field plate electrodes **13** is formed in an endless shape (square ring shape) surrounding the emitter routing portion **23** in the plan view. At least one of the plural field plate electrodes **13** may be formed in an ended shape.

[0057] The equipotential electrode **14** is formed in the scribe region **9**. The equipotential electrode **14** is drawn around in a band shape along the field plate electrodes **13**. In this embodiment, the equipotential electrode **14** is formed in an endless shape (square ring shape) surrounding the field plate electrodes **13** in the plan view. Thereby, the equipotential electrode **14** is formed as an EQR

(Equi-potential Ring) electrode.

[0058] FIG. 2 is a sectional view taken along line II-II shown in FIG. 1. FIG. 3 is a sectional view taken along line III-III shown in FIG. 1.

[0059] With reference to FIGS. 2 and 3, the semiconductor layer 2 has a single layer structure including an n.sup.- type semiconductor substrate 31 in this embodiment. The n.sup.- type semiconductor substrate 31 is made of a silicon FZ substrate formed by the FZ (Floating Zone) method. A collector electrode 32 serving as a rear surface electrode is connected to the second principal surface 4 of the semiconductor layer 2.

[0060] In the device formation region 6, in a surface layer portion of the first principal surface 3 of the semiconductor layer 2, a p type channel region 33 is formed. In this embodiment, the p type channel region 33 is formed in a square shape having four sides parallel to the side surfaces 5a to 5d of the semiconductor layer 2 in the plan view. The p type channel region 33 is formed to be spaced from the peripheral edge of the semiconductor layer 2 to the inner region of the semiconductor layer 2. In this embodiment, the device formation region 6 is defined by the p type channel region 33.

[0061] In the device formation region 6, in a region on the second principal surface 4 side of the semiconductor layer 2 with respect to the p type channel region 33, an n.sup.- type drift region 34 is formed. The n.sup.- type drift region 34 is electrically connected to the p type channel region 33.

[0062] The n.sup.- type drift region 34 is formed by utilizing a part of the n.sup.- type semiconductor substrate 31. The n.sup.- type drift region 34 is formed in the outer region 7, the withstand voltage retention region 8, and the scribe region 9 in addition to the device formation region 6.

[0063] In the device formation region 6, in a surface layer portion of the second principal surface 4 of the semiconductor layer 2, a p.sup.+ type collector region 35 is formed. The p.sup.+ type collector region 35 is electrically connected to the collector electrode 32 on the second principal surface 4 of the semiconductor layer 2. The p.sup.+ type collector region 35 is electrically connected to the n' type drift region 34 in the semiconductor layer 2.

[0064] The p.sup.+ type collector region 35 is formed in the outer region 7, the withstand voltage retention region 8, and the scribe region 9 in addition to the device formation region 6. In this embodiment, the p.sup.+ type collector region 35 is formed over the entire surface layer portion of the second principal surface 4 of the semiconductor layer 2.

[0065] The p.sup.+ type collector region 35 is formed in a non-facing region not facing the p type channel region 33 in addition to a facing region facing the p type channel region 33 on the second principal surface 4 of the semiconductor layer 2.

[0066] In a region between the n.sup.- type drift region 34 and the p.sup.+ type collector region 35, an n type buffer region 36 is formed in this embodiment. The n type buffer region 36 is a high-concentration and low-resistant region having an n type impurity concentration higher than an n type impurity concentration of the n.sup.- type drift region 34. The n type buffer region 36 extends in the region between the n.sup.- type drift region 34 and the p.sup.+ type collector region 35 along the second principal surface 4 of the semiconductor layer 2.

[0067] In this embodiment, the p.sup.+ type collector region 35 is electrically connected to the n.sup.- type drift region 34 via the n type buffer region 36. Although not shown in the figures, an n type field stop region may be formed in the region between the n type buffer region 36 and the p.sup.+ type collector region 35.

[0068] In the device formation region 6, on the first principal surface 3 of the semiconductor layer 2, plural trench gate structures 37 are formed to be spaced from each other. In this embodiment, the plural trench gate structures 37 are formed in a band shape extending along an arbitrary one direction in the plan view. Trench gate structures 37 extending in a grid shape in the plan view may be formed.

[0069] Each of the trench gate structures 37 includes a gate trench 38, a gate insulating film 39, and

an embedded gate electrode **40**.

[0070] The gate trench **38** is formed on the first principal surface **3** of the semiconductor layer **2**.

[0071] The gate trench **38** passes through the p type channel region **33**, and has a bottom wall positioned in the n.sup.- type drift region **34**.

[0072] The gate insulating film **39** is formed in a film shape along an inner wall surface of the gate trench **38**. One surface (front surface on the semiconductor layer **2** side) and the other surface of the gate insulating film **39** are formed along the inner wall surface of the gate trench **38**.

[0073] The gate trench **38** may be formed in a square shape having side walls which extend substantially perpendicularly to the first principal surface **3** of the semiconductor layer **2**. The gate trench **38** may be formed in a tapered shape whose opening area is larger than a bottom area. The bottom wall of the gate trench **38** may be formed in parallel to the first principal surface **3** of the semiconductor layer **2**. The bottom wall of the gate trench **38** may be formed in a projected and curved shape extending outward.

[0074] The embedded gate electrode **40** is embedded in the gate trench **38** with the gate insulating film **39** being placed between the electrode and the gate trench. The inner wall surface of the gate trench **38** forms a part of the first principal surface **3** of the semiconductor layer **2**. The embedded gate electrode **40** may include at least one of polysilicon, copper, aluminum, or tungsten.

[0075] A surface insulating film **41** is formed on the first principal surface **3** of the semiconductor layer **2**. The gate insulating film **39** continues to the surface insulating film **41** outside the gate trench **38**.

[0076] The trench gate structures **37** include a trench gate structure **37** formed in a region of the device formation region **6** close to the outer region **7**. The embedded gate electrode **40** of this trench gate structure **37** includes a gate lead portion **42** drawn out from the inside of the gate trench **38** toward the outer region **7**.

[0077] The gate lead portion **42** is drawn out to a region immediately below the gate pad **16** (not shown) and/or a region immediately below the gate finger **17**. FIG. **3** shows an example in which the gate lead portion **42** is drawn out to the region immediately below the gate finger **17**.

[0078] The gate lead portion **42** is electrically connected to the gate pad **16** and the gate finger **17**. Thereby, the embedded gate electrode **40** is electrically connected to the gate electrode **11** via the gate lead portion **42**.

[0079] On the side of each of the trench gate structures **37**, an n.sup.+ type emitter region **43** is formed in a surface layer portion of the p type channel region **33**. The n.sup.+ type emitter region **43** is exposed from the first principal surface **3** of the semiconductor layer **2**.

[0080] On the side of each of the trench gate structures **37**, the n.sup.+ type emitter region **43**, the p type channel region **33**, and the n.sup.- type drift region **34** are formed in this order from the first principal surface **3** of the semiconductor layer **2** toward the second principal surface **4**.

[0081] In a region between the adjacently-placed trench gate structures **37**, a first contact recessed portion **44** for the emitter pad **22** is formed in the surface layer portion of the p type channel region **33**. The first contact recessed portion **44** extends in a band shape along the trench gate structures **37**. In a case where the trench gate structures **37** formed in a grid shape in the plan view are formed, the first contact recessed portion **44** may be formed in a region surrounded by the trench gate structures **37**.

[0082] The first contact recessed portion **44** is formed by digging into the first principal surface **3** of the semiconductor layer **2**. The first contact recessed portion **44** is shallower than the gate trench **38**. A bottom wall of the first contact recessed portion **44** is positioned in the p type channel region **33**. The p type channel region **33** is exposed from a side wall and the bottom wall of the first contact recessed portion **44**. The n.sup.+ type emitter region **43** is exposed from the side wall of the first contact recessed portion **44**.

[0083] In this embodiment, the p type channel region **33** includes a p.sup.+ type contact region **45** whose p type impurity concentration is higher than that of the other region. The p.sup.+ type

contact region **45** is formed in a region of the p type channel region **33** along the bottom wall of the first contact recessed portion **44**.

[0084] The p.sup.+ type contact region **45** is formed along the side wall of the first contact recessed portion **44**. The p.sup.+ type contact region **45** is also formed along corner portions connecting the bottom wall and the side wall of the first contact recessed portion **44**. The p.sup.+ type contact region **45** may be formed along only the bottom wall of the first contact recessed portion **44**.

[0085] The p type channel region **33** is shared by the adjacently-placed trench gate structures **37**. The embedded gate electrode **40** faces the n.sup.+ type emitter region **43**, the p type channel region **33**, and the n.sup.- type drift region **34** across the gate insulating film **39**.

[0086] An IGBT channel structure **46** is formed by the trench gate structures **37**, the n.sup.+ type emitter region **43**, the p type channel region **33**, and the n.sup.- type drift region **34**. An IGBT channel is formed in a region of the p type channel region **33** between the n.sup.+ type emitter region **43** and the n.sup.- type drift region **34**.

[0087] A dummy channel structure **47** is formed in the device formation region **6** and the region of the outer region **7** in the device formation region **6**. The dummy channel structure **47** has a structure similar to the channel structure **46** except the point in which the n.sup.+ type emitter region **43** is not included. In the dummy channel structure **47**, no IGBT channel is formed.

[0088] In the region of the device formation region **6** and the outer region **7**, an electric current in the device formation region **6** and an electric current outside the device formation region **6** are merged. Thus, there is a tendency in which current density becomes relatively high. In the IGBT channel structure **46**, structurally, the npn type parasitic bipolar transistor described above is formed.

[0089] Therefore, when the IGBT channel structure **46** is formed in the region of the device formation region **6** and the outer region **7**, the npn type parasitic bipolar transistor is turned on to possibly cause a latch-up.

[0090] Meanwhile, the dummy channel structure **47** does not include any n.sup.+ type emitter region **43**. Thus, structurally, no npn type parasitic bipolar transistor is formed. Therefore, a latch-up due to the parasitic bipolar transistor can be suppressed by the dummy channel structure **47**.

[0091] In the outer region **7**, a p.sup.+ type terminal region **51** is formed in the surface layer portion of the first principal surface **3** of the semiconductor layer **2**. The p.sup.+ type terminal region **51** defines the outer region **7**. The device formation region **6** is also defined by a region surrounded by an inner peripheral edge of the p.sup.+ type terminal region **51**.

[0092] The p.sup.+ type terminal region **51** is a high-concentration and low-resistant region having a p type impurity concentration higher than a p type impurity concentration of the p type channel region **33**. In this embodiment, the p.sup.+ type terminal region **51** is formed in an endless shape (square ring shape) surrounding the p type channel region **33** in the plan view.

[0093] A bottom portion of the p.sup.+ type terminal region **51** is positioned in the second principal surface **4** side of the semiconductor layer **2** with respect to a bottom portion of the p type channel region **33** regarding the thickness direction of the semiconductor layer **2**. The bottom portion of the p.sup.+ type terminal region **51** is positioned in the second principal surface **4** side of the semiconductor layer **2** with respect to the bottom wall of the gate trench **38** regarding the thickness direction of the semiconductor layer **2**. The bottom portion of the p.sup.+ type terminal region **51** is positioned in the first principal surface **3** side of the semiconductor layer **2** with respect to the p.sup.+ type collector region **35**.

[0094] An inner edge region of the p.sup.+ type terminal region **51** positioned in the p type channel region **33** side overlaps the bottom wall of the gate trench **38** of the dummy channel structure **47**. The inner edge region of the p.sup.+ type terminal region **51** may overlap the bottom portion of the p type channel region **33**.

[0095] The p.sup.+ type terminal region **51** faces the p.sup.+ type collector region **35** across a part of the n.sup.- type drift region **34**. The p.sup.+ type terminal region **51** faces the gate pad **16** and/or

the gate finger 17 across the surface insulating film 41. The p.sup.+ type terminal region 51 faces the gate lead portion 42 across the surface insulating film 41.

[0096] In the withstand voltage retention region 8, p.sup.+ type field limit regions 52 are formed in the surface layer portion of the first principal surface 3 of the semiconductor layer 2. The p.sup.+ type field limit regions 52 relax the electric field in the withstand voltage retention region 8.

[0097] The p.sup.+ type field limit regions 52 have a p type impurity concentration substantially equal to the p type impurity concentration of the p.sup.+ type terminal region 51. The p.sup.+ type field limit regions 52 have a depth substantially equal to a depth of the p.sup.+ type terminal region 51.

[0098] The p.sup.+ type field limit regions 52 are formed in a region between the outer region 7 and the scribe region 9 along the p.sup.+ type terminal region 51. In this embodiment, the p.sup.+ type field limit regions 52 are formed in an endless shape (square ring shape) surrounding the p.sup.+ type terminal region 51 in the plan view. Thereby, the p.sup.+ type field limit regions 52 are formed as FLR (Field Limiting Ring) regions.

[0099] In this embodiment, the p.sup.+ type field limit regions 52 include plural (five in this embodiment) p.sup.+ type field limit regions 52 formed to be spaced from each other in this order from the outer region 7 toward the scribe region 9. At least one p.sup.+ type field limit region 52 may be formed. Five or more p.sup.+ type field limit regions 52 may be formed.

[0100] In the scribe region 9, an n.sup.+ type channel stop region 53 is formed in the surface layer portion of the first principal surface 3 of the semiconductor layer 2. The n.sup.+ type channel stop region 53 is a high-concentration and low-resistant region having an n type impurity concentration higher than the n type impurity concentration of the n.sup.- type drift region 34. The n.sup.+ type channel stop region 53 suppresses spread of a depletion layer from a pn junction portion formed in the inner region of the semiconductor layer 2.

[0101] The n.sup.+ type channel stop region 53 is formed along the p.sup.+ type field limit regions 52. The n.sup.+ type channel stop region 53 is formed in an endless shape (square ring shape) surrounding the p.sup.+ type field limit regions 52 in the plan view. The n.sup.+ type channel stop region 53 may cross a region between the withstand voltage retention region 8 and the scribe region 9.

[0102] In the outer region 7, a second contact recessed portion 54 for the emitter pad 22 and a third contact recessed portion 55 for the emitter routing portion 23 are formed in a surface layer portion of the p.sup.+ type terminal region 51.

[0103] The second contact recessed portion 54 is formed at the first principal surface 3 of the semiconductor layer 2. The second contact recessed portion 54 is formed along an inner edge of the p.sup.+ type terminal region 51. The second contact recessed portion 54 is formed in an endless shape (square ring shape) surrounding the device formation region 6 in the plan view.

[0104] The second contact recessed portion 54 has a bottom wall positioned in the p.sup.+ type terminal region 51. The second contact recessed portion 54 has a depth substantially equal to a depth of the first contact recessed portion 44.

[0105] The third contact recessed portion 55 is formed on the first principal surface 3 of the semiconductor layer 2. The third contact recessed portion 55 is formed along an outer edge of the p.sup.+ type terminal region 51. The third contact recessed portion 55 is formed in an endless shape (square ring shape) surrounding the device formation region 6 in the plan view.

[0106] The third contact recessed portion 55 has a bottom wall positioned in the p.sup.+ type terminal region 51. The third contact recessed portion 55 has a depth substantially equal to the depth of the first contact recessed portion 44.

[0107] The p.sup.+ type terminal region 51 includes a p.sup.+ type contact region 56 formed in a region along the bottom wall of the second contact recessed portion 54. The p.sup.+ type contact region 56 is a high-concentration region having a p type impurity concentration higher than those of the other regions in the p.sup.+ type terminal region 51.

[0108] The p.sup.+ type contact region **56** is also formed along a side wall of the second contact recessed portion **54**. The p.sup.+ type contact region **56** is also formed along corner portions connecting the bottom wall and the side wall of the second contact recessed portion **54**. The p.sup.+ type contact region **56** may be formed along only the bottom wall of the second contact recessed portion **54**.

[0109] The p.sup.+ type terminal region **51** includes a p.sup.+ type contact region **57** formed in a region along the bottom wall of the third contact recessed portion **55**. The p.sup.+ type contact region **57** is a high-concentration region having a p type impurity concentration higher than those of the other regions in the p.sup.+ type terminal region **51**.

[0110] The p.sup.+ type contact region **57** is also formed along a side wall of the third contact recessed portion **55**. The p.sup.+ type contact region **57** is also formed along corner portions connecting the bottom wall and the side wall of the third contact recessed portion **55**. The p.sup.+ type contact region **57** may be formed along only the bottom wall of the third contact recessed portion **55**.

[0111] In the withstand voltage retention region **8**, a fourth contact recessed portion **58** for each of the field plate electrodes **13** is formed in a surface layer portion of each of the p.sup.+ type field limit regions **52**.

[0112] The fourth contact recessed portions **58** are formed at the first principal surface **3** of the semiconductor layer **2**. The fourth contact recessed portions **58** are formed along the p.sup.+ type field limit regions **52**. The fourth contact recessed portions **58** are formed in an endless shape (square ring shape) surrounding the device formation region **6** in the plan view.

[0113] Each of the fourth contact recessed portions **58** has a bottom wall positioned in each of the p.sup.+ type field limit regions **52**. The fourth contact recessed portions **58** have a depth substantially equal to the depth of the first contact recessed portion **44**.

[0114] Each of the p.sup.+ type field limit regions **52** includes a p.sup.+ type contact region **59** formed in a region along a bottom wall of each of the fourth contact recessed portions **58**. Each of the p.sup.+ type contact regions **59** is a high-concentration region having a p type impurity concentration higher than those of the other regions in each of the p.sup.+ type field limit regions **52**.

[0115] Each of the p.sup.+ type contact regions **59** is formed along, in addition to the bottom wall of the fourth contact recessed portion **58**, the side wall of the fourth contact recessed portion **58**, and corner portions connecting the bottom wall and the side wall of the fourth contact recessed portions **58**. Each of the p.sup.+ type contact regions **59** may be formed along only the bottom wall of the fourth contact recessed portion **58**.

[0116] In the scribe region **9**, a fifth contact recessed portion **60** for the equipotential electrode **14** is formed in the surface layer portion of the first principal surface **3** of the semiconductor layer **2**.

[0117] The fifth contact recessed portion **60** is formed by digging into the first principal surface **3** of the semiconductor layer **2**. The fifth contact recessed portion **60** is formed along the n.sup.+ type channel stop region **53**. The fifth contact recessed portion **60** is formed in an endless shape (square ring shape) surrounding the withstand voltage retention region **8** in the plan view.

[0118] The fifth contact recessed portion **60** has a depth substantially equal to the depth of the first contact recessed portion **44**. The n.sup.+ type channel stop region **53** is exposed from the fifth contact recessed portion **60**.

[0119] In the semiconductor layer **2**, a p.sup.+ type contact region **61** is formed in a region along a bottom wall of the fifth contact recessed portion **60**. The p.sup.+ type contact region **61** may be formed along a side wall of the fifth contact recessed portion **60**.

[0120] The p.sup.+ type contact region **61** may be formed along corner portions connecting the bottom wall and the side wall of the fifth contact recessed portion **60**. The p.sup.+ type contact region **61** may be formed along only the bottom wall of the fifth contact recessed portion **60**.

[0121] An insulating layer **64** is formed on the first principal surface **3** of the semiconductor layer

2. The insulating layer **64** covers the device formation region **6**, the outer region **7**, the withstand voltage retention region **8**, and the scribe region **9**. The insulating layer **64** may have a laminated structure in which plural insulating films are laminated. The insulating layer **64** may have a single layer structure including a single insulating layer. The insulating layer **64** may include an oxide film (SiO₂ film) or a nitride film (SiN film).

[0122] First contact holes **65**, a second contact hole **67**, and a third contact hole **68** are formed in the insulating layer **64**. The first contact holes **65**, the second contact hole **67**, and the third contact hole **68** are respectively formed for the emitter electrode **12**.

[0123] The first contact holes **65**, the second contact hole **67**, and the third contact hole **68** are respectively flush with and communicate with the first contact recessed portions **44**, the second contact recessed portion **54**, and the third contact recessed portion **55** in a one-to-one relationship.

[0124] Fourth contact holes **69** for the field plate electrodes **13** are formed in the insulating layer **64**. The fourth contact holes **69** are flush with and communicate with the fourth contact recessed portions **58**.

[0125] A fifth contact hole **70** for the equipotential electrode **14** is formed in the insulating layer **64**. The fifth contact hole **70** is flush with and communicates with the fifth contact recessed portion **60**.

[0126] A gate contact hole **71** for the gate electrode **11** is formed in the insulating layer **64**. From the gate contact hole **71**, the gate lead portion **42** is exposed in the region immediately below the gate pad **16** and/or the region immediately below the gate finger **17**. FIG. **3** shows an example in which the gate contact hole **71** is formed in the region immediately below the gate finger **17**.

[0127] The front surface electrode **10** is formed on the insulating layer **64**. The front surface electrode **10** has a laminated structure including a first conductor layer **72** formed on the insulating layer **64** and a second conductor layer **73** formed on the first conductor layer **72**.

[0128] One surface (front surface on the semiconductor layer **2** side) and the other surface of the first conductor layer **72** are formed along a front surface of the insulating layer **64** and a front surface of the semiconductor layer **2**. One surface (front surface on the semiconductor layer **2** side) and the other surface of the second conductor layer **73** are formed along the first conductor layer **72**.

[0129] The first conductor layer **72** may have a single layer structure including a titanium nitride layer or a titanium layer. The first conductor layer **72** may have a laminated structure including a titanium nitride layer and a titanium layer formed on the titanium nitride layer. The first conductor layer **72** functions as a barrier electrode layer by including the titanium nitride layer and/or the titanium layer.

[0130] The second conductor layer **73** may include at least one of copper, aluminum, an alloy including copper, or an alloy including aluminum. The second conductor layer **73** may include an aluminum-copper alloy (Al—Cu alloy). The second conductor layer **73** may include an aluminum-silicon-copper alloy (Al—Si—Cu alloy).

[0131] The gate pad **16** and/or the gate finger **17** come into the gate contact hole **71** from above the insulating layer **64**. The gate pad **16** and/or the gate finger **17** are electrically connected to the gate lead portion **42** in the gate contact hole **71**.

[0132] The emitter pad **22** comes into the first contact holes **65** and the second contact hole **67** from above the insulating layer **64**. The emitter pad **22** is electrically connected to the p type channel region **33** and the n.sup.+ type emitter region **43** in the first contact holes **65**. The emitter pad **22** is electrically connected to the p.sup.+ type terminal region **51** in the second contact hole **67**.

[0133] The emitter routing portion **23** comes into the third contact hole **68** from above the insulating layer **64**. The emitter routing portion **23** is electrically connected to the p.sup.+ type terminal region **51** in the third contact hole **68**. Although not shown in the figures, the emitter connecting portion **24** may come into the second contact hole **67**.

[0134] In the semiconductor layer **2**, the avalanche current flowing in the region outside the device formation region **6** is absorbed by the emitter electrode **12** via the p.sup.+ type terminal region **51**.

Therefore, the avalanche current absorbing structure **25** further includes, in addition to the emitter pad **22**, the emitter routing portion **23**, and the emitter connecting portion **24**, the p.sup.+ type terminal region **51** electrically connected to these.

[0135] The field plate electrodes **13** come into the fourth contact holes **69** from above the insulating layer **64**. The field plate electrodes **13** are electrically connected to the p.sup.+ type field limit regions **52** in the fourth contact holes **69**.

[0136] The equipotential electrode **14** comes into the fifth contact hole **70** from above the insulating layer **64**. The equipotential electrode **14** is electrically connected to the n.sup.+ type channel stop region **53** in the fifth contact hole **70**.

[0137] FIG. **4** is a plan view for explaining a structure on the second principal surface **4** side of the semiconductor layer **2** shown in FIG. **1**.

[0138] With reference to FIG. **4**, an n.sup.+ type cathode region **80** is formed in the surface layer portion of the second principal surface **4** of the semiconductor layer **2**. In FIG. **4**, the n.sup.+ type cathode region **80** is shown by cross-hatching. With reference to FIGS. **2** to **4**, the n.sup.+ type cathode region **80** is formed by selectively introducing n type impurities to the second principal surface **4** of the semiconductor layer **2** (p.sup.+ type collector region **35**).

[0139] The n.sup.+ type cathode region **80** has an n type impurity concentration higher than an n type impurity concentration of the semiconductor layer **2**. The n.sup.+ type cathode region **80** is a region containing n type impurities and p type impurities, in which the n type impurity concentration of the n type impurities is higher than a p type impurity concentration of the p type impurities.

[0140] The n.sup.+ type cathode region **80** includes an n.sup.+ type inner cathode region **81** and an n.sup.+ type outer cathode region **82**. The n.sup.+ type inner cathode region **81** is formed in the device formation region **6**. The n.sup.+ type outer cathode region **82** is formed in the outer region **7**.

[0141] The n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82** are formed through an n type impurity implantation process utilizing the same ion implantation mask. The n.sup.+ type outer cathode region **82** has a depth substantially equal to a depth of the n.sup.+ type inner cathode region **81**. The n.sup.+ type outer cathode region **82** has an n type impurity concentration substantially equal to an n type impurity concentration of the n.sup.+ type inner cathode region **81**.

[0142] The n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82** may be formed through an n type impurity implantation process utilizing different ion implantation masks. The n.sup.+ type outer cathode region **82** may be formed with the depth different from the depth of the n.sup.+ type inner cathode region **81**. The n.sup.+ type outer cathode region **82** may be formed with an n type impurity concentration different from the n type impurity concentration of the n.sup.+ type inner cathode region **81**.

[0143] On the second principal surface **4** of the semiconductor layer **2**, a free space region **83** (see the region surrounded by a double chain line) and a cathode formation region **84** are set in the device formation region **6**. The free space region **83** is a region where the n.sup.+ type inner cathode region **81** is not formed. The cathode formation region **84** is a region where the n.sup.+ type inner cathode region **81** is formed.

[0144] In this embodiment, the free space region **83** is set in a region overlapping the gate pad **16** in the plan view. The free space region **83** may enclose a region where the gate pad **16** overlaps the device formation region **6** in the plan view. The free space region **83** may be set in a square shape in the plan view.

[0145] The cathode formation region **84** is set in a region overlapping the emitter pad **22** in the plan view. In this embodiment, the cathode formation region **84** is set in a C shape in a region outside the free space region **83** in the plan view.

[0146] The n.sup.+ type outer cathode region **82** is formed in the inner region from the dummy

channel structure **47** in the cathode formation region **84**. The n.sup.+ type outer cathode region **82** is preferably not formed in a region immediately below the dummy channel structure **47**.
[0147] With this structure, in the vicinity of the region of the device formation region **6** and the outer region **7**, an electric current path passing through the n.sup.+ type outer cathode region **82** is not formed. Therefore, in the region of the device formation region **6** and the outer region **7**, an increase in an undesired current and an undesired electric field concentration due to the increase in the current can be suppressed. As a matter of course, a structure in which the n.sup.+ type outer cathode region **82** is formed in the region immediately below the dummy channel structure **47** may be adopted.

[0148] Since the n.sup.+ type inner cathode region **81** is not formed in the free space region **83**, a pattern becomes non-uniform in the device formation region **6**. More specifically, the n.sup.+ type inner cathode region **81** is formed in a non-uniform pattern which is at least non-symmetric with respect to a point in the device formation region **6**.

[0149] The n.sup.+ type inner cathode region **81** may be formed in a non-uniform pattern which is non-symmetric with respect to a line in the device formation region **6**. The n.sup.+ type inner cathode region **81** may be formed in a non-uniform pattern which is non-symmetric with respect to a point and with respect to a line in the device formation region **6**.

[0150] Hereinafter, the +X direction, the -X direction, the +Y direction, and the -Y direction shown in FIG. **4** may be used for convenience of description. The +X direction and the -X direction are two directions oriented along the pair of side surfaces **5c**, **5d** and will be simplified as the "X direction" when being referred to collectively. The +Y direction and the -Y direction are two directions oriented along the pair of side surfaces **5a**, **5b** and will be simplified as the "Y direction" when being referred to collectively.

[0151] In this embodiment, the n.sup.+ type inner cathode region **81** includes a line-shaped pattern portion **85** of a meandering form drawn around so as to continuously extend in the cathode formation region **84** in the plan view. The line-shaped pattern portion **85** includes plural first lines **86** and plural second lines **87**.

[0152] The plural first lines **86** extend along the X direction and are spaced from each other along the Y direction. The plural second lines **87** extend along the Y direction and connect the first lines **86** adjacent to each other in the Y direction.

[0153] The plural first lines **86** include a first line group **88**, a second line group **89**, and a third line group **90**. The first line group **88** includes the plural first lines **86** formed in a region on the +Y direction side from the free space region **83** in the device formation region **6**. The second line group **89** includes the plural first lines **86** formed in a region on the -Y direction side from the free space region **83** in the device formation region **6**.

[0154] The third line group **90** includes the plural first lines **86** formed in a region between the first line group **88** and the second line group **89** in the device formation region **6**. The third line group **90** is formed in a region on the +X direction side from the free space region **83**.

[0155] In this embodiment, an X direction length of the second line group **89** is substantially equal to an X direction length of the first line group **88**. The X direction length of the second line group **89** may be set as a value different from the X direction length of the first line group **88**. An X direction length of the third line group **90** is set as a value smaller than the X direction length of the first line group **88** and the X direction length of the second line group **89**.

[0156] The first line group **88** includes a first line drawing out portion **91** drawn out to a region on the +Y direction side from the free space region **83** so as to face the free space region **83** in the Y direction. The second line group **89** includes a second line drawing out portion **92** drawn out to a region on the -Y direction side from the free space region **83** so as to face the first line group **88** in the Y direction across the free space region **83**.

[0157] The second lines **87** include second lines **87A** and second lines **87B**. Each of the second lines **87A** connects +X direction end portions of two first lines **86** adjacent to each other in the Y

direction. Each of the second lines **87B** connects -X direction end portions of two first lines **86** adjacent to each other in the Y direction. The second lines **87A** and the second lines **87B** are formed one after the other along the Y direction.

[0158] A line width of the n.sup.+ type inner cathode region **81** may be not less than 1 μm and not more than 100 μm . The line width of the n.sup.+ type inner cathode region **81** may be not less than 10 μm and not more than 50 μm . The line width of the n.sup.+ type inner cathode region **81** is defined by a Y direction width of the first lines **86** and an X direction width of the second lines **87**.

[0159] The line width of the n.sup.+ type inner cathode region **81** may be uniform or non-uniform. The n.sup.+ type inner cathode region **81** may include plural first lines **86** having respectively different Y direction widths. The n.sup.+ type inner cathode region **81** may include plural second lines **87** having respectively different X direction widths.

[0160] A ratio SB/SA of a planar area SB of the free space region **83** with respect to a planar area SA of the device formation region **6** may be not less than 0.03 (3%) and not more than 0.3 (30%). A ratio SC/SA of a planar area SC of the n.sup.+ type inner cathode region **81** with respect to the planar area SA of the device formation region **6** is set as a value smaller than a ratio SD/SA of a planar area SD of the p.sup.+ type collector region **35** with respect to the planar area SA of the device formation region **6**.

[0161] Hereinafter, the ratio SC/SA will be referred to as area ratio SC/SA of the n.sup.+ type inner cathode region **81**. The ratio SD/SA will be referred to as area ratio SD/SA of the p.sup.+ type collector region **35**. The area ratio SC/SA of the n.sup.+ type inner cathode region **81** may be not more than 0.1 (10%). The area ratio SC/SA of the n.sup.+ type inner cathode region **81** may be not less than 0.01 (1%) and not more than 0.07 (7%).

[0162] With reference to FIG. 4, the n.sup.+ type outer cathode region **82** is formed in the outer region **7**. The n.sup.+ type outer cathode region **82** is formed along a periphery of the device formation region **6** in the outer region **7**. The n.sup.+ type outer cathode region **82** is formed in the non-facing region outside the facing region to face the p type channel region **33** in the surface layer portion of the second principal surface **4** of the semiconductor layer **2**.

[0163] In this embodiment, the n.sup.+ type outer cathode region **82** includes line-shaped pattern portions **93** drawn around so as to continuously extend along the periphery of the device formation region **6**. The n.sup.+ type outer cathode region **82** has a structure in which the four line-shaped pattern portions **93** formed along four sides of the device formation region **6** are integrated. Thereby, the n.sup.+ type outer cathode region **82** is formed in an endless shape (square ring shape) surrounding the device formation region **6** in the plan view.

[0164] With reference to FIG. 3 again, the n.sup.+ type outer cathode region **82** is formed to pass through the p.sup.+ type collector region **35** in the outer region **7**. The n.sup.+ type outer cathode region **82** is electrically connected to a region of the semiconductor layer **2** (in this embodiment, to the n.sup.- type drift region **34** and the n type buffer region **36**).

[0165] The n.sup.+ type outer cathode region **82** faces the p.sup.+ type terminal region **51** across a region of the semiconductor layer **2** (in this embodiment, to the n.sup.- type drift region **34** and the n type buffer region **36**). In this embodiment, the entire n.sup.+ type outer cathode region **82** faces the p.sup.+ type terminal region **51**.

[0166] A line width of the n.sup.+ type outer cathode region **82** may be not less than 1 μm and not more than 100 μm . The line width of the n.sup.+ type outer cathode region **82** may be not less than 10 μm and not more than 50 μm . The line width of the n.sup.+ type outer cathode region **82** is a width in the direction orthogonal to the direction in which the n.sup.+ type outer cathode region **82** extends.

[0167] Next, an electrical structure of the semiconductor device **1** will be described. FIG. 5 is a circuit diagram showing the electrical structure of the semiconductor device **1** shown in FIG. 1.

[0168] With reference to FIG. 5, the semiconductor device **1** includes the IGBT and a free wheeling diode D. An RC-IGBT (a Reverse Conducting-Insulated Gate Bipolar Transistor) is formed by the

IGBT and the free wheeling diode D.

[0169] The IGBT includes the gate electrode **11**, the emitter electrode **12**, and the collector electrode **32**. The free wheeling diode D is connected between the emitter electrode **12** and the collector electrode **32**.

[0170] The free wheeling diode D includes an anode and a cathode. The anode of the free wheeling diode D is electrically connected to the emitter electrode **12** of the IGBT. The cathode of the free wheeling diode D is electrically connected to the collector electrode **32** of the IGBT.

[0171] In this embodiment, the free wheeling diode D includes an inner free wheeling diode Di and an outer free wheeling diode Do.

[0172] The inner free wheeling diode Di includes a pn junction portion formed in a region between the p type channel region **33** and the n.sup.- type drift region **34** in the device formation region **6**. The inner free wheeling diode Di includes the n.sup.+ type inner cathode region **81** as the cathode and the p type channel region **33** as the anode.

[0173] An anode electrode of the inner free wheeling diode Di is formed mainly by the emitter pad **22** of the emitter electrode **12**. A cathode electrode of the inner free wheeling diode Di is formed by the collector electrode **32** connected to the n.sup.+ type inner cathode region **81**.

[0174] The outer free wheeling diode Do includes a pn junction portion formed in a region between the p.sup.+ type terminal region **51** and the n.sup.- type drift region **34** in the outer region **7**. The outer free wheeling diode Do includes the n.sup.+ type outer cathode region **82** as the cathode and the p.sup.+ type terminal region **51** as the anode.

[0175] An anode electrode of the outer free wheeling diode Do is formed mainly by the emitter routing portion **23** of the emitter electrode **12**. A cathode electrode of the outer free wheeling diode Do is formed by the collector electrode **32** connected to the n.sup.+ type outer cathode region **82**.

[0176] With the semiconductor device **1**, forward current IF-forward voltage VF characteristics described by using FIG. **6** can be achieved. FIG. **6** is a graph in which the forward current IF-forward voltage VF characteristics of the semiconductor device **1** shown in FIG. **1** are obtained by simulation. In FIG. **6**, the vertical axis indicates the forward current IF, and the horizontal axis indicates the forward voltage VF.

[0177] The forward voltage VF is a voltage applied between the emitter electrode **12** and the collector electrode **32**. The forward current IF is an electric current flowing between the emitter electrode **12** and the collector electrode **32** when the forward voltage VF is applied.

[0178] The forward voltage VF is also a voltage applied between the n.sup.+ type emitter region **43** and the p.sup.+ type collector region **35**. The forward current IF is also an electric current flowing between the n.sup.+ type emitter region **43** and the p.sup.+ type collector region **35**.

[0179] FIG. **6** shows a first characteristic S1 (see the broken line) and a second characteristic S2 (see the solid line). The first characteristic S1 is a forward current IF-forward voltage VF characteristics of a semiconductor device having no n.sup.+ type outer cathode region **82** (hereinafter, simply referred to as “semiconductor device according to the first reference example”). The second characteristic S2 is the forward current IF-forward voltage VF characteristics of the semiconductor device **1**.

[0180] The first characteristic S1 includes a first characteristic S1A determined by a first simulation condition and a second characteristic S1B determined by a second simulation condition. The second characteristic S2 includes a first characteristic S2A determined by the first simulation condition and a second characteristic S2B determined by the second simulation condition.

[0181] The first simulation condition is a condition under which the forward voltage VF of not less than 0 V and not more than 5 V is applied in a state where a gate-emitter voltage VGE is 0 V, and the forward current IF is measured. The gate-emitter voltage VGE is a voltage applied between the gate electrode **11** and the emitter electrode **12** (n.sup.+ type emitter region **43**).

[0182] The second simulation condition is a condition under which the forward voltage VF of not less than 0 V and not more than 5 V is applied in a state where the gate-emitter voltage VGE is 15

V, and the forward current IF is measured.

[0183] With reference to the first characteristic S1A, in the semiconductor device according to the first reference example, the forward current IF-forward voltage VF characteristics by the first simulation condition are normal. However, with reference to the second characteristic S1B, by the second simulation condition, even when the forward voltage VF is increased from 0 V to 5 V, the forward current IF does not exceed 10 A in the semiconductor device according to the first reference example.

[0184] It is found that when the forward voltage VF is further applied in a state where the gate-emitter voltage VGE is applied, it is difficult for the forward current IF to flow in the semiconductor device according to the first reference example. Therefore, it is found that, in order to obtain a relatively high forward current IF, a relatively high forward voltage VF should be applied in the semiconductor device according to the first reference example. Such a problem can result in a snap-back phenomenon.

[0185] Meanwhile, with reference to the first characteristic S2A, in the semiconductor device **1**, the forward current IF-forward voltage VF characteristics by the first simulation condition are normal. With reference to the second characteristic S2B, by the second simulation condition, the forward current IF-forward voltage VF characteristics of the semiconductor device **1** are improved from the forward current IF-forward voltage VF characteristics of the semiconductor device according to the reference example. Thus, it is found that large variation of the forward current IF-forward voltage VF characteristics due to application/non-application of the gate-emitter voltage VGE is suppressed in the semiconductor device **1**.

[0186] In the semiconductor device **1**, in a case where the forward voltage VF of 1.6 V is applied in a state where the gate-emitter voltage VGE is 15 V, the forward current is not less than 2 A. In a case where the forward voltage VF of 2 V is applied in a state where the gate-emitter voltage VGE is 15 V, the forward current is not less than 8 A. In a case where the forward voltage VF of 3 V is applied in a state where the gate-emitter voltage VGE is 15 V, the forward current IF is not less than 20 A.

[0187] As described above, with the semiconductor device **1**, the IGBT and the free wheeling diode D are formed in the device formation region **6**. The free wheeling diode D includes the inner free wheeling diode Di and the outer free wheeling diode Do.

[0188] The inner free wheeling diode Di includes the pn junction portion formed in the region between the p type channel region **33** and the n.sup.- type drift region **34** in the device formation region **6**. The outer free wheeling diode Do includes the pn junction portion formed in the region between the p.sup.+ type terminal region **51** and the n.sup.- type drift region **34** in the outer region **7**.

[0189] The inner free wheeling diode Di includes the n.sup.+ type inner cathode region **81** as the cathode and the p type channel region **33** as the anode. The anode electrode of the inner free wheeling diode Di is formed mainly by the emitter pad **22** of the emitter electrode **12**. The cathode electrode of the inner free wheeling diode Di is formed by the collector electrode **32** connected to the n.sup.+ type inner cathode region **81**.

[0190] The outer free wheeling diode Do includes the n.sup.+ type outer cathode region **82** as the cathode and the p.sup.+ type terminal region **51** as the anode. The anode electrode of the outer free wheeling diode Do is formed mainly by the emitter routing portion **23** of the emitter electrode **12**. The cathode electrode of the outer free wheeling diode Do is formed by the collector electrode **32** connected to the n.sup.+ type outer cathode region **82**.

[0191] With such a structure, even when the forward voltage VF is applied in a state where the gate-emitter voltage VGE is applied, difficulty for the forward current IF to flow can be suppressed (see the second characteristic S2 of FIG. **6**). Therefore, deterioration of the forward voltage VF can be suppressed, so that a snap-back phenomenon due to the deterioration of the forward voltage VF can be suppressed.

[0192] In the semiconductor device **1**, the n.sup.+ type inner cathode region **81** of the inner free wheeling diode Di includes the line-shaped pattern portion **85** drawn around so as to continuously and non-uniformly extend in the device formation region **6**. With such a structure, the following effect to be described with FIG. **7** can be exerted.

[0193] FIG. **7** is a graph in which a peak forward surge current IFSM of the semiconductor device **1** shown in FIG. **1** is obtained by simulation. In FIG. **7**, the vertical axis indicates the peak forward surge current IFSM, and the horizontal axis indicates the area ratio SC/SA of the n.sup.+ type inner cathode region **81**.

[0194] The peak forward surge current IFSM is the peak value of a commercial sinusoidal current of not less than one cycle that is tolerated within a range in which the semiconductor device **1** does not break down. The higher the peak forward surge current IFSM, the better the resistance of the semiconductor device **1** against the peak forward surge current IFSM (hereinafter, simply referred to as “peak forward surge current resistance”).

[0195] FIG. **7** shows a plotted point P1. The plotted point P1 shows characteristics of the peak forward surge current IFSM of the semiconductor device **1**. The plotted point P1 shows characteristics of the peak forward surge current IFSM in a case where the area ratio SC/SA of the n.sup.+ type inner cathode region **81** is 0.037 (3.7%).

[0196] FIG. **7** also shows plotted points P2, P3, P4, and P5. The plotted points P2 to P5 show characteristics of the peak forward surge current IFSM of a semiconductor device according to a second reference example. The plotted points P2 to P5 are connected by an approximation line L.

[0197] The semiconductor device according to the second reference example does not include the n.sup.+ type outer cathode region **82**. The n.sup.+ type inner cathode region **81** of the semiconductor device according to the second reference example is formed in a regular arrays-of-dots pattern in the device formation region **6**.

[0198] The plotted point P2 shows characteristics of the peak forward surge current IFSM in a case where the area ratio SC/SA of the n.sup.+ type inner cathode region **81** is 0.012 (1.2%). The plotted point P3 shows characteristics of the peak forward surge current IFSM in a case where the area ratio SC/SA of the n.sup.+ type inner cathode region **81** is 0.019 (1.9%).

[0199] The plotted point P4 shows characteristics of the peak forward surge current IFSM in a case where the area ratio SC/SA of the n.sup.+ type inner cathode region **81** is 0.023 (2.4%). The plotted point P5 shows characteristics of the peak forward surge current IFSM in a case where the area ratio SC/SA of the n.sup.+ type inner cathode region **81** is 0.032 (3.2%).

[0200] With reference to the approximation line L, when the area ratio SC/SA of the n.sup.+ type inner cathode region **81** becomes smaller, the peak forward surge current IFSM is lowered. When the area ratio SC/SA of the n.sup.+ type inner cathode region **81** becomes larger, the peak forward surge current IFSM is increased. Therefore, in the semiconductor device according to the second reference example, a generally linear relationship is established between the planar area SC and the peak forward surge current IFSM.

[0201] However, in the semiconductor device according to the second reference example, even in a case where the area ratio SC/SA of the n.sup.+ type inner cathode region **81** is any value, the peak forward surge current IFSM is a relatively low value (less than 700 A).

[0202] With reference to the approximation line L, in the semiconductor device according to the second reference example, it is considered that when the area ratio SC/SA of the n.sup.+ type inner cathode region **81** comes close to “one,” a relatively high value of the peak forward surge current IFSM can be realized.

[0203] However, in reality, as the area ratio SC/SA of the n.sup.+ type inner cathode region **81** comes closer to “one,” the area ratio SD/SA of the p.sup.+ type collector region **35** comes closer to “zero.” Thus, the function of the IGBT is lost.

[0204] Therefore, in the semiconductor device according to the second reference example, even when the area ratio SC/SA of the n.sup.+ type inner cathode region **81** is adjusted, the peak forward

surge current IFSM can be adjusted only within the linear relationship shown by the approximation line L at the end.

[0205] Such a linear relationship is particularly notable in a case where plural n.sup.+ type inner cathode regions **81** formed in a square shape, a circular shape, etc., in the plan view are arranged in regular arrays in the device formation region **6**.

[0206] Thus, the semiconductor device **1** adopts a structure where the n.sup.+ type inner cathode region **81** includes the line-shaped pattern portion **85** continuously and non-uniformly extending in the device formation region **6**.

[0207] Thereby, the peak forward surge current IFSM can be set separately from the linear relationship shown by the approximation line L (see the plotted point P1 of FIG. 7). More specifically, with the semiconductor device **1**, the peak forward surge current IFSM of not less than 700 A (not less than 900 A in FIG. 7) can be realized.

[0208] As described above, it is possible to provide the semiconductor device **1** with which the deterioration of the forward voltage VF can be suppressed by including the n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82**. Thus, it is possible to provide the semiconductor device **1** with which the peak forward surge current resistance can be improved by the n.sup.+ type inner cathode region **81** including the continuously and non-uniformly extending line-shaped pattern portion **85**.

[0209] The structure of the n.sup.+ type outer cathode region **82** and the structure of the n.sup.+ type inner cathode region **81** are not limited to the preferred embodiment described above but various other modes are available. Hereinafter, modification examples of the n.sup.+ type outer cathode region **82** and modification examples of the n.sup.+ type inner cathode region **81** will be described.

[0210] FIG. 8 is a plan view showing a first modification example of the n.sup.+ type outer cathode region **82**. FIG. 8 is a plan view corresponding to FIG. 4. In FIG. 8, structures corresponding to the structures shown in FIG. 4, etc., will be provided with the same reference symbols and description thereof will be omitted.

[0211] In FIG. 8, for convenience of description, only the n.sup.+ type outer cathode region **82** is shown and the structures, etc., within the device formation region **6** are omitted. In FIG. 8, the n.sup.+ type outer cathode region **82** is shown by cross-hatching.

[0212] The n.sup.+ type outer cathode region **82** according to the present modification example has an ended-shaped structure in which an arbitrary one of the four line-shaped pattern portions **93** is formed as a separate body from the other line-shaped pattern portions **93**.

[0213] The n.sup.+ type outer cathode region **82** may have a structure in which two or three of the four line-shaped pattern portions **93** are formed as separate bodies from the other line-shaped pattern portions **93**.

[0214] The n.sup.+ type outer cathode region **82** may have a structure in which at least one of the four line-shaped pattern portions **93** is formed as a separate body from the other line-shaped pattern portions **93**.

[0215] The n.sup.+ type outer cathode region **82** may have a structure in which at least two adjacent line-shaped pattern portions **93** among the four line-shaped pattern portions **93** are connected to each other.

[0216] With the semiconductor device **1** having the n.sup.+ type outer cathode region **82** of such a structure, the operations and the effects which are substantially similar to the operations and the effects described above can also be exerted.

[0217] FIG. 9 is a plan view showing a second modification example of the n.sup.+ type outer cathode region **82**. FIG. 9 is a plan view corresponding to FIG. 4. In FIG. 9, structures corresponding to the structures shown in FIG. 4, etc., will be provided with the same reference symbols and description thereof will be omitted.

[0218] In FIG. 9, for convenience of description, only the n.sup.+ type outer cathode region **82** is

shown and the structures, etc., within the device formation region **6** are omitted. In FIG. **9**, the n.sup.+ type outer cathode region **82** is shown by cross-hatching.

[0219] The n.sup.+ type outer cathode region **82** according to the present modification example has plural line-shaped pattern portions **93** which are formed as separated bodies from each other. The plural line-shaped pattern portions **93** are respectively formed along two sides facing each other among four sides of the device formation region **6**.

[0220] The n.sup.+ type outer cathode region **82** may include two line-shaped pattern portions **93** formed along only arbitrary two sides crossing each other among four sides of the device formation region **6**. The n.sup.+ type outer cathode region **82** may include one line-shaped pattern portion **93** formed along only arbitrary one side among four sides of the device formation region **6**.

[0221] The n.sup.+ type outer cathode region **82** may include three line-shaped pattern portions **93** formed along arbitrary three sides among four sides of the device formation region **6**. The n.sup.+ type outer cathode region **82** may include a line-shaped pattern portion **93** formed along at least one side among four sides of the device formation region **6**.

[0222] With the semiconductor device **1** having the n.sup.+ type outer cathode region **82** of such a structure, the operations and the effects which are substantially similar to the operations and the effects described above can also be exerted.

[0223] FIG. **10** is a plan view showing a third modification example of the n.sup.+ type outer cathode region **82**. FIG. **10** is a plan view corresponding to FIG. **4**. In FIG. **10**, structures corresponding to the structures shown in FIG. **4**, etc., will be provided with the same reference symbols and description thereof will be omitted.

[0224] In FIG. **10**, for convenience of description, only the n.sup.+ type outer cathode regions **82** are shown and the structures, etc., within the device formation region **6** are omitted. In FIG. **10**, the n.sup.+ type outer cathode regions **82** are shown by cross-hatching.

[0225] In the present modification example, the plural n.sup.+ type outer cathode regions **82** are formed. The plural n.sup.+ type outer cathode regions **82** include a first n.sup.+ type outer cathode region **82A**, a second n.sup.+ type outer cathode region **82B**, and a third n.sup.+ type outer cathode region **82C**. The first to third n.sup.+ type outer cathode regions **82A**, **82B**, **82C** are formed to be spaced from each other in this order from the outer region **7** toward the withstand voltage retention region **8**.

[0226] The first to third n.sup.+ type outer cathode regions **82A**, **82B**, **82C** may be formed in the outer region **7**. At least one of the first to third n.sup.+ type outer cathode regions **82A**, **82B**, **82C** may be formed in the withstand voltage retention region **8**. The third n.sup.+ type outer cathode region **82C** may be formed in the withstand voltage retention region **8**.

[0227] The first n.sup.+ type outer cathode region **82A** includes a line-shaped pattern portion **93** drawn around so as to continuously extend along the periphery of the device formation region **6**. The first n.sup.+ type outer cathode region **82A** is formed in an endless shape (square ring shape) surrounding the device formation region **6** in the plan view.

[0228] The second n.sup.+ type outer cathode region **82B** includes a line-shaped pattern portion **93** drawn around so as to continuously extend along a periphery of the first n.sup.+ type outer cathode region **82A**. The second n.sup.+ type outer cathode region **82B** is formed in an endless shape (square ring shape) surrounding the first n.sup.+ type outer cathode region **82A** in the plan view.

[0229] The third n.sup.+ type outer cathode region **82C** includes a line-shaped pattern portion **93** drawn around so as to continuously extend along a periphery of the second n.sup.+ type outer cathode region **82B**. The third n.sup.+ type outer cathode region **82C** is formed in an endless shape (square ring shape) surrounding the second n.sup.+ type outer cathode region **82B** in the plan view.

[0230] With the semiconductor device **1** having the n.sup.+ type outer cathode regions **82** of such a structure, the operations and the effects which are substantially similar to the operations and the effects described above can also be exerted. In the present modification example, the number of the n.sup.+ type outer cathode regions **82** is not limited to three. Two n.sup.+ type outer cathode

regions **82** may enclose the device formation region **6**. Four or more n.sup.+ type outer cathode regions **82** may enclose the device formation region **6**.

[0231] FIG. **11** is a plan view showing a fourth modification example of the n.sup.+ type outer cathode region **82**. FIG. **11** is a plan view corresponding to FIG. **4**. In FIG. **11**, structures corresponding to the structures shown in FIG. **4**, etc., will be provided with the same reference symbols and description thereof will be omitted.

[0232] In FIG. **11**, for convenience of description, only the n.sup.+ type outer cathode region **82** is shown and the structures, etc., within the device formation region **6** are omitted. In FIG. **11**, the n.sup.+ type outer cathode region **82** is shown by cross-hatching.

[0233] The n.sup.+ type outer cathode region **82** according to the present modification example includes plural dot-shaped pattern portions **94**. The plural dot-shaped pattern portions **94** are formed to be spaced from each other along the periphery of the device formation region **6**.

[0234] In the present modification example, the n.sup.+ type outer cathode region **82** includes the dot-shaped pattern portions **94** formed in a circular shape in the plan view. The n.sup.+ type outer cathode region **82** may include dot-shaped pattern portions **94** formed in a polygonal shape such as a triangle, a square, and a hexagon in the plan view in place of, or in addition to the circular dot-shaped pattern portions **94**. The n.sup.+ type outer cathode region **82** may include dot-shaped pattern portions **94**, etc., formed in an oval shape in the plan view in place of, or in addition to the circular dot-shaped pattern portions **94**.

[0235] With the semiconductor device **1** having the n.sup.+ type outer cathode region **82** of such a structure, the operations and the effects which are substantially similar to the operations and the effects described above can also be exerted.

[0236] The n.sup.+ type outer cathode regions **82** according to the first preferred embodiment described above and the first to fourth modification examples described above may be arbitrarily combined with each other.

[0237] For example, the dot-shaped pattern portions **94** according to the fourth modification example may be combined with the first to third modification examples. In this case, the n.sup.+ type outer cathode region **82** includes the line-shaped pattern portions **93** drawn around so as to continuously extend along the periphery of the device formation region **6**, and the dot-shaped pattern portions **94** formed to be spaced from each other along the periphery of the device formation region **6**.

[0238] FIG. **12** is a plan view showing a first modification example of the n.sup.+ type inner cathode region **81**. FIG. **12** is a plan view corresponding to FIG. **4**. In FIG. **12**, structures corresponding to the structures shown in FIG. **4**, etc., will be provided with the same reference symbols and description thereof will be omitted.

[0239] FIG. **12** shows an example in which the n.sup.+ type outer cathode regions **82** according to the third modification example described above (see FIG. **10**) are applied. In FIG. **12**, the n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode regions **82** are shown by cross-hatching.

[0240] The n.sup.+ type inner cathode region **81** according to the present modification example has a structure in which the first line group **88** does not include the first line drawing out portion **91** and the second line group **89** does not include the second line drawing out portion **92**.

[0241] The n.sup.+ type inner cathode region **81** according to the present modification example is unevenly placed on the +X direction side of the device formation region **6**. The n.sup.+ type inner cathode region **81** is formed in a pattern non-uniform (non-symmetric with respect to a point) with respect to the device formation region **6**.

[0242] The n.sup.+ type inner cathode region **81** may have a structure in which the first line group **88** includes the first line drawing out portion **91** whereas the second line group **89** does not include the second line drawing out portion **92**. The n.sup.+ type inner cathode region **81** may have a structure in which the first line group **88** does not include the first line drawing out portion **91**

whereas the second line group **89** includes the second line drawing out portion **92**.

[0243] With the semiconductor device **1** having the n.sup.+ type inner cathode region **81** of such a structure, the operations and the effects which are substantially similar to the operations and the effects described above can also be exerted.

[0244] FIG. **13** is a plan view showing a second modification example of the n.sup.+ type inner cathode region **81**. FIG. **13** is a plan view corresponding to FIG. **4**. In FIG. **13**, structures corresponding to the structures shown in FIG. **4**, etc., will be provided with the same reference symbols and description thereof will be omitted.

[0245] FIG. **13** shows an example in which n.sup.+ type outer cathode regions **82** formed by combining the n.sup.+ type outer cathode region **82** according to the second modification example (see FIG. **9**) and the n.sup.+ type outer cathode region **82** according to the fourth modification example (see FIG. **11**) with the n.sup.+ type outer cathode regions **82** according to the third modification example (see FIG. **10**) are applied. In FIG. **13**, the n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode regions **82** are shown by cross-hatching.

[0246] The n.sup.+ type inner cathode region **81** according to the present modification example includes comb-shaped line-shaped pattern portions **85** continuing to each other in the plan view. More specifically, in the n.sup.+ type inner cathode region **81** according to the present modification example, the second lines **87** include only the second lines **87A** each of which connects the +X direction end portions of the first lines **86** adjacent to each other in the Y direction. Thereby, the n.sup.+ type inner cathode region **81** is formed to be non-uniform (non-symmetric with respect to a point) with respect to the device formation region **6**.

[0247] With the semiconductor device **1** of such a structure, the operations and the effects which are substantially similar to the operations and the effects of the first preferred embodiment described above can also be exerted.

[0248] FIG. **14** is a plan view showing a second modification example of the n.sup.+ type inner cathode region **81**. FIG. **14** is a plan view corresponding to FIG. **4**. In FIG. **14**, structures corresponding to the structures shown in FIG. **4**, etc., will be provided with the same reference symbols and description thereof will be omitted.

[0249] FIG. **14** shows an example in which the n.sup.+ type outer cathode region **82** according to the fourth modification example described above (see FIG. **11**) is applied. In FIG. **14**, the n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82** are shown by cross-hatching.

[0250] The n.sup.+ type inner cathode region **81** according to the present modification example includes plural dot-shaped pattern portions **95**. The plural dot-shaped pattern portions **95** are arranged in arrays to be spaced from each other in the X direction and the Y direction in the cathode formation region **84** formed by excluding the free space region **83** from the device formation region **6**.

[0251] The n.sup.+ type inner cathode region **81** may include plural dot-shaped pattern portions **95** arranged in a zigzag form in place of, or in addition to the plural dot-shaped pattern portions **95** arranged in arrays. The n.sup.+ type inner cathode region **81** is formed in a pattern non-uniform (non-symmetric with respect to a point) with respect to the device formation region **6**.

[0252] In the present modification example, the n.sup.+ type inner cathode region **81** includes the dot-shaped pattern portions **95** formed in a circular shape in the plan view. The n.sup.+ type inner cathode region **81** may include dot-shaped pattern portions **95** formed in a polygonal shape such as a triangle, a square, and a hexagon in the plan view in place of, or in addition to the circular dot-shaped pattern portions **95**. The n.sup.+ type inner cathode region **81** may include dot-shaped pattern portions **95**, etc., formed in an oval shape in the plan view in place of, or in addition to the circular dot-shaped pattern portions **95**.

[0253] In the semiconductor device **1** of such a structure, it is considered to be difficult to set the peak forward surge current IFSM separately from the linear relationship established between the

planar area SC of the n.sup.+ type inner cathode region **81** and the peak forward surge current IFSM (see the approximation line L of FIG. 7). However, since the n.sup.+ type outer cathode region **82** is formed, the deterioration of the forward voltage VF can be suppressed.

[0254] With the present modification example, combination of the n.sup.+ type inner cathode region **81** including the line-shaped pattern portion **85** and/or the dot-shaped pattern portions **95**, and the n.sup.+ type outer cathode region **82** including the line-shaped pattern portions **93** and/or the dot-shaped pattern portions **94** is allowed. Thereby, an advantage that the forward voltage VF and the peak forward surge current IFSM can be set without providing an unnecessary restriction on degree of freedom of design can be obtained.

[0255] FIG. **15** is a plan view showing a fourth modification example of the n.sup.+ type inner cathode region **81**. FIG. **15** is a plan view corresponding to FIG. **4**. In FIG. **15**, structures corresponding to the structures shown in FIG. **4**, etc., will be provided with the same reference symbols and description thereof will be omitted.

[0256] FIG. **15** shows an example in which the n.sup.+ type outer cathode region **82** according to the second modification example described above (see FIG. **9**) is applied. In FIG. **15**, the n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82** are shown by cross-hatching.

[0257] With reference to FIG. **15**, in the present modification example, the gate pad **16** is formed in a central region of the device formation region **6** in the plan view, and the gate finger **17** is formed in the outer region **7**. The gate pad **16** is connected to the gate finger **17** via a connecting portion **96**. The connecting portion **96** is drawn out from the gate pad **16** toward the outer region **7**.

[0258] The emitter pad **22** is formed in a C-shaped region partitioned by the peripheral edge of the gate pad **16**, the peripheral edge of the gate finger **17**, and a peripheral edge of the connecting portion **96** in the plan view. In this embodiment, the emitter pad **22** is formed in a C shape formed along an inner edge of the gate pad **16** and an inner edge of the gate finger **17** in the plan view.

[0259] In the present modification example, on the second principal surface **4** of the semiconductor layer **2**, the free space region **83** is set in a region immediately below the gate pad **16** formed in the central region of the device formation region **6**. The cathode formation region **84** is set in a region between a peripheral edge of the free space region **83** and the peripheral edge of the device formation region **6**.

[0260] The n.sup.+ type inner cathode region **81** according to the present modification example includes a line-shaped pattern portion **85** formed in a square-spiral shape continuously extending to enclose the free space region **83** in the plan view. The line-shaped pattern portion **85** according to the present modification example includes plural first lines **86** and plural second lines **87** as in the case of the first preferred embodiment described above.

[0261] The n.sup.+ type inner cathode region **81** is formed in a region excluding the central region of the device formation region **6**. With such a structure, the n.sup.+ type inner cathode region **81** is formed in a pattern non-uniform (non-symmetric with respect to a point) with respect to the device formation region **6**.

[0262] With the semiconductor device **1** having such a structure, the operations and the effects which are substantially similar to the operations and the effects described above can also be exerted.

[0263] In the present modification example, the example in which the n.sup.+ type inner cathode region **81** includes the line-shaped pattern portion **85** formed in a square-spiral shape in the plan view is described. However, the n.sup.+ type inner cathode region **81** may include a line-shaped pattern portion **85** formed in a circular-spiral shape in the plan view. The n.sup.+ type inner cathode region **81** may include a line-shaped pattern portion **85** formed in a polygonal-spiral shape which is a non-square-spiral shape such as a triangle-, or hexagon-spiral shape in the plan view.

[0264] The n.sup.+ type inner cathode regions **81** according to the first preferred embodiment described above and the first to fourth modification examples described above may be arbitrarily

combined with each other.

[0265] For example, the n.sup.+ type inner cathode region **81** including the dot-shaped pattern portions **95** according to the fourth modification example (see FIG. **15**) may be combined with a part of the n.sup.+ type inner cathode region **81** including the line-shaped pattern portion **85** of a meandering form according to the first preferred embodiment. In this case, the n.sup.+ type inner cathode region **81** has a structure including the line-shaped pattern portion **85** of a meandering form and the dot-shaped pattern portions **95**.

[0266] For example, the comb-shaped n.sup.+ type inner cathode region **81** according to the second modification example (see FIG. **13**) may be combined with a part of the n.sup.+ type inner cathode region **81** of a meandering form according to the first preferred embodiment. In this case, the n.sup.+ type inner cathode region **81** according to the first preferred embodiment has a structure including the line-shaped pattern portion **85** of a meandering form and the comb-shaped line-shaped pattern portions **85**. As a matter of course, the n.sup.+ type inner cathode region **81** of a structure including the line-shaped pattern portion **85** of a meandering form, the comb-shaped line-shaped pattern portions **85**, and the dot-shaped pattern portions **95** may be adopted.

[0267] FIG. **16** is a plan view showing a modification example of the n.sup.+ type cathode region **80**. FIG. **16** is a plan view corresponding to FIG. **4**. In FIG. **16**, structures corresponding to the structures shown in FIG. **4**, etc., will be provided with the same reference symbols and description thereof will be omitted. In FIG. **16**, the n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82** are shown by cross-hatching.

[0268] In the present modification example, the n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82** are integrated. The figure shows an example in which the n.sup.+ type inner cathode region **81** of a meandering form and the n.sup.+ type outer cathode region **82** formed in a square ring shape in the plan view are integrated.

[0269] A portion of the n.sup.+ type inner cathode region **81** drawn out as a part of the n.sup.+ type outer cathode region **82** forms a part of the outer free wheeling diode Do in the outer region **7**. A portion of the n.sup.+ type outer cathode region **82** drawn out as a part of the n.sup.+ type inner cathode region **81** forms a part of the inner free wheeling diode Di in the device formation region **6**.

[0270] With the semiconductor device **1** having such a structure, the operations and the effects which are substantially similar to the operations and the effects described above can also be exerted.

[0271] In an arbitrary combination structure of the n.sup.+ type outer cathode regions **82** according to the first to fourth modification examples (see FIGS. **8** to **11**) and the n.sup.+ type inner cathode regions **81** according to the first to fourth modification examples (see FIGS. **12** to **15**), a structure in which the n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82** are integrated may be adopted.

[0272] For example, the n.sup.+ type inner cathode region **81** may include a spiral line-shaped pattern portion **85** continuously drawn around from the n.sup.+ type outer cathode region **82** toward the central region of the device formation region **6**.

[0273] The n.sup.+ type inner cathode region **81** may include a comb-shaped line-shaped pattern portion **85** continuously drawn around from the n.sup.+ type outer cathode region **82** toward the central region of the device formation region **6**.

[0274] FIG. **17** is a schematic sectional view of a semiconductor device **101** according to a second preferred embodiment. Hereinafter, structures corresponding to the structures described for the semiconductor device **1** will be provided with the same reference symbols and description thereof will be omitted.

[0275] The semiconductor device **1** has the trench gate type IGBT. Meanwhile, the semiconductor device **101** has a planar gate type IGBT. The semiconductor device **101** includes a semiconductor layer **2**. P type channel regions **33** are formed to be spaced from each other in a surface layer portion of a first principal surface **3** of the semiconductor layer **2**.

[0276] In a surface layer portion of each of the p type channel regions **33**, an n.sup.+ type emitter region **43** and a p.sup.+ type contact region **45** are formed to be spaced from a peripheral edge of the p type channel region **33** to the inside. In a surface layer portion of a second principal surface **4** of the semiconductor layer **2**, an n type buffer region **36**, a p.sup.+ type collector region **35**, and an n.sup.+ type cathode region **80** are formed. In the semiconductor layer **2**, a region between the p type channel region **33** and the n type buffer region **36** is the n.sup.– type drift region **34** described above.

[0277] In a device formation region **6**, a planar gate structure **102** is formed on the first principal surface **3** of the semiconductor layer **2**. The planar gate structure **102** includes a gate insulating film **103** and a gate electrode **104**.

[0278] The gate electrode **104** faces the n.sup.+ type emitter region **43**, the p type channel regions **33**, and the n' type drift region **34** across the gate insulating film **103**. The planar gate structure **102** is covered with the insulating layer **64** described above. In this embodiment, the device formation region **6** is defined by a region surrounded by a p.sup.+ type terminal region **51**.

[0279] A contact hole **105** from which the p type channel regions **33** and the n.sup.+ type emitter region **43** are exposed is formed in the insulating layer **64**. The emitter electrode **12** comes into the contact hole **105** from above the insulating layer **64**.

[0280] The emitter electrode **12** is electrically connected to the p type channel regions **33** and the n.sup.+ type emitter region **43** in the contact hole **105**. A collector electrode **32** is formed on the second principal surface **4** side of the semiconductor layer **2**. The collector electrode **32** is electrically connected to the p.sup.+ type collector region **35** and the n.sup.+ type cathode region **80**.

[0281] As described above, with the semiconductor device **101**, the operations and the effects which are similar to the operations and the effects described for the semiconductor device **1** can also be exerted. The structure of the semiconductor device **101** may be assembled into the first preferred embodiment and each of the modification examples. That is, in the first preferred embodiment and each of the modification examples, in place of, or in addition to the trench gate type IGBT, the planar gate type IGBT may be formed in the device formation region **6**.

[0282] Although the preferred embodiments of the present invention are described above, the present invention can further be implemented in other modes.

[0283] For example, in each of the preferred embodiments described above and each of the modification examples described above, the semiconductor layer **2** may include, in place of the n.sup.– type semiconductor substrate **31**, a p type semiconductor substrate and an n.sup.– type epitaxial layer formed on the p type semiconductor substrate.

[0284] Both the p type semiconductor substrate and the n.sup.– type epitaxial layer may be made of silicon. The n.sup.– type epitaxial layer is formed by epitaxial growth of silicon on a principal surface of the p type semiconductor substrate.

[0285] In this case, the p type semiconductor substrate corresponds to the p.sup.+ type collector region **35**. The n type epitaxial layer corresponds to the n.sup.– type drift region **34**. The n.sup.+ type inner cathode region **81** and the n.sup.+ type outer cathode region **82** are formed by implanting n type impurities to the p type semiconductor substrate (p.sup.+ type collector region **35**).

[0286] In each of the preferred embodiments described above and each of the modification examples described above, the gate electrode **11** may be formed in a region other than the central region of the side surface **5a**. In this case, the gate electrode **11** may be formed along a corner portion connecting arbitrary two side surfaces among the side surfaces **5a**, **5b**, **5c**, **5d** in the plan view.

[0287] In each of the preferred embodiments described above and each of the modification examples described above, the pair of open ends **18**, **19** of the gate finger **17** may be formed along a region other than the side surface **5b**. In this case, the pair of open ends **18**, **19** of the gate finger

17 may be formed along a corner portion positioned opposing the corner portion where the gate electrode **11** is arranged.

[0288] In each of the preferred embodiments described above and each of the modification examples described above, a structure where no p.sup.+ type terminal region **51** is provided may be adopted. In this case, the outer free wheeling diode Do includes the pn junction portion formed in the region between the p type channel region **33** and the n.sup.- type drift region **34**. The outer free wheeling diode Do includes the n.sup.+ type outer cathode region **82** as the cathode and the p type channel region **33** as the anode.

[0289] In each of the preferred embodiments described above and each of the modification examples described above, a structure where no emitter routing portion **23** and no emitter connecting portion **24** are provided may be adopted. In this case, the anode electrode of the outer free wheeling diode Do is formed by the emitter pad **22**. The cathode electrode of the outer free wheeling diode Do is formed by the collector electrode **32** connected to the n.sup.+ type outer cathode region **82**.

[0290] In each of the preferred embodiments described above and each of the modification examples described above, the n.sup.+ type inner cathode region **81** may be formed in a stripe shape in the plan view. In each of the preferred embodiments described above and each of the modification examples described above, the n.sup.+ type inner cathode region **81** may be formed in a grid shape in the plan view.

[0291] In each of the preferred embodiments described above and each of the modification examples described above, the free space region **83** may be set in a region outside the region overlapping the gate pad **16** in the plan view.

[0292] In each of the preferred embodiments described above and each of the modification examples described above, the free space region **83** may be not overlapping the gate pad **16** in the plan view.

[0293] In each of the preferred embodiments described above and each of the modification examples described above, a structure where no free space region **83** is provided may be adopted.

[0294] In each of the preferred embodiments described above and each of the modification examples described above, the n.sup.+ type inner cathode region **81** may be formed in a region overlapping the gate pad **16** in the plan view.

[0295] In each of the preferred embodiments described above and each of the modification examples described above, the n.sup.+ type inner cathode region **81** may include line-shaped pattern portions **85** in uniform arrangement (symmetric with respect to a line and to a point) in the plan view in the device formation region **6**.

[0296] In each of the preferred embodiments described above and each of the modification examples described above, the n.sup.+ type inner cathode region **81** may include dot-shaped pattern portions **95** in uniform arrangement (symmetric with respect to a line and to a point) in the plan view in the device formation region **6**.

[0297] In each of the preferred embodiments described above and each of the modification examples described above, at least a part of the n.sup.+ type outer cathode region **82** may be formed in the withstand voltage retention region **8**. In this case, at least a part of the n.sup.+ type outer cathode region **82** may face the p.sup.+ type field limit regions **52**.

[0298] In each of the preferred embodiments described above and each of the modification examples described above, all of the n.sup.+ type outer cathode region **82** may be formed in a withstand voltage retention region **8**. In this case, at least a part of or all of the n.sup.+ type outer cathode region **82** may face the p.sup.+ type field limit regions **52**.

[0299] In each of the preferred embodiments described above and each of the modification examples described above, at least a part of the n.sup.+ type outer cathode region **82** may be formed in the scribe region **9**. In this case, at least a part of the n.sup.+ type outer cathode region **82** may face the n.sup.+ type channel stop region **53**.

[0300] In each of the preferred embodiments described above and each of the modification examples described above, all of the n.sup.+ type outer cathode region **82** may be formed in the scribe region **9**. In this case, at least a part of or all of the n.sup.+ type outer cathode region **82** may face the n.sup.+ type channel stop region **53**.

[0301] In each of the preferred embodiments described above and each of the modification examples described above, an arrangement in which the conductivity types of the respective semiconductor portions are inverted may be adopted. That is, a p type portion may be of an n type and an n type portion may be of a p type.

[0302] While preferred embodiments of the present invention have been described in detail above, these are merely specific examples used to clarify the technical contents of the present invention, and the present invention should not be interpreted as being limited only to these specific examples, and the scope of the present invention shall be limited only by the appended claims.

Claims

1. A semiconductor device comprising: a semiconductor layer of a first conductivity type that has a first principal surface on one side and a second principal surface on another side, and that has a recessed portion recessed toward the second principal surface side at a peripheral portion of the first principal surface; a device formation region that is provided in the semiconductor layer; an outer region that is provided outside the device formation region in the semiconductor layer; a channel region of a second conductivity type that is formed in a surface layer portion of the first principal surface at the device formation region; an emitter region of the first conductivity type that is formed in a surface layer portion of the channel region; an embedded gate electrode that is embedded in the first principal surface across a gate insulating film at the device formation region and faces the channel region across the gate insulating film; a collector region of the second conductivity type that is formed in a surface layer portion of the second principal surface at the device formation region; dot-shaped impurity regions of the first conductivity type that are formed in the collector region, and that are each formed in a circular shape in a plan view; field limit regions that are formed in the surface layer portion of the first principal surface at the outer region; an outer impurity region of the first conductivity type that is formed along a bottom surface and a side surface of the recessed portion in the surface layer portion of the first principal surface at the outer region so as to form a boundary with the semiconductor layer; a gate pad that is electrically connected to the embedded gate electrode on the first principal surface side, and that has an inner edge which is arranged closer to a center of the device formation region than the outer region in the plan view; an Al-based wiring layer that is electrically connected to one of the field limit regions on the first principal surface side; and a recessed conductive layer that extends over both the recessed portion and the first principal surface on the first principal surface side, and that has a lower surface with a slope obliquely inclined with respect to the first principal surface due to the recessed portion; wherein the Al-based wiring layer has an inner end positioned on the field limit region side, and an outer end extending toward the peripheral portion side of the first principal surface in a cross sectional view, the field limit region has an impurity inner end positioned on a center side of the first principal surface, an impurity outer end positioned on the peripheral portion side of the first principal surface, and an impurity center positioned in an intermediate portion between the impurity inner end and the impurity outer end in the cross sectional view, a length between the impurity center of the field limit region and the outer end of the Al-based wiring layer is greater than a length between the impurity center of the field limit region and the inner end of the Al-based wiring layer, the inner end of the Al-based wiring layer is positioned on the impurity center side of the field limit region with respect to the impurity inner end of the field limit region, the impurity outer end of the field limit region is positioned on the impurity center side of the field limit region with respect to the outer end of the Al-based wiring layer, and the dot-shaped impurity regions

include the dot-shaped impurity regions arranged closer to the center of the device formation region than the inner edge of the gate pad in the plan view.

2. The semiconductor device according to claim 1, wherein the recessed portion is formed continuously over an entire circumference of the peripheral portion of the semiconductor layer.

3. The semiconductor device according to claim 1, wherein the Al-based wiring layer is formed continuously along the peripheral portion of the first principal surface.

4. The semiconductor device according to claim 1, wherein the dot-shaped impurity regions are formed in an arrays-of-dots pattern.

5. The semiconductor device according to claim 1, wherein the first conductivity type is n-type, and the second conductivity type is p-type.

6. The semiconductor device according to claim 1, further comprising: a drift region of the first conductivity type that is formed in the semiconductor layer; wherein an IGBT (Insulated Gate Bipolar Transistor) channel structure is formed by the embedded gate electrode, the emitter region of the first conductivity type, the channel region of the second conductivity type, and the drift region of the first conductivity type, and the gate pad overlaps with a portion of the device formation region and a portion of the outer region.

7. The semiconductor device according to claim 1, further comprising: an emitter electrode that is connected to the emitter region on the first principal surface side; and a collector electrode that is connected to the collector region and the dot-shaped impurity regions on the second principal surface side.

8. The semiconductor device according to claim 1, further comprising: an insulating layer that covers the first principal surface in the outer region and has a sloped surface obliquely inclined with respect to the first principal surface due to the recessed portion; wherein the recessed conductive layer has a covering portion that covers the sloped surface of the insulating layer, and the lower surface of the covering portion of the recessed conductive layer has the slope extending along the sloped surface of the insulating layer.
